Data & AI

13. Language Models

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Last Time

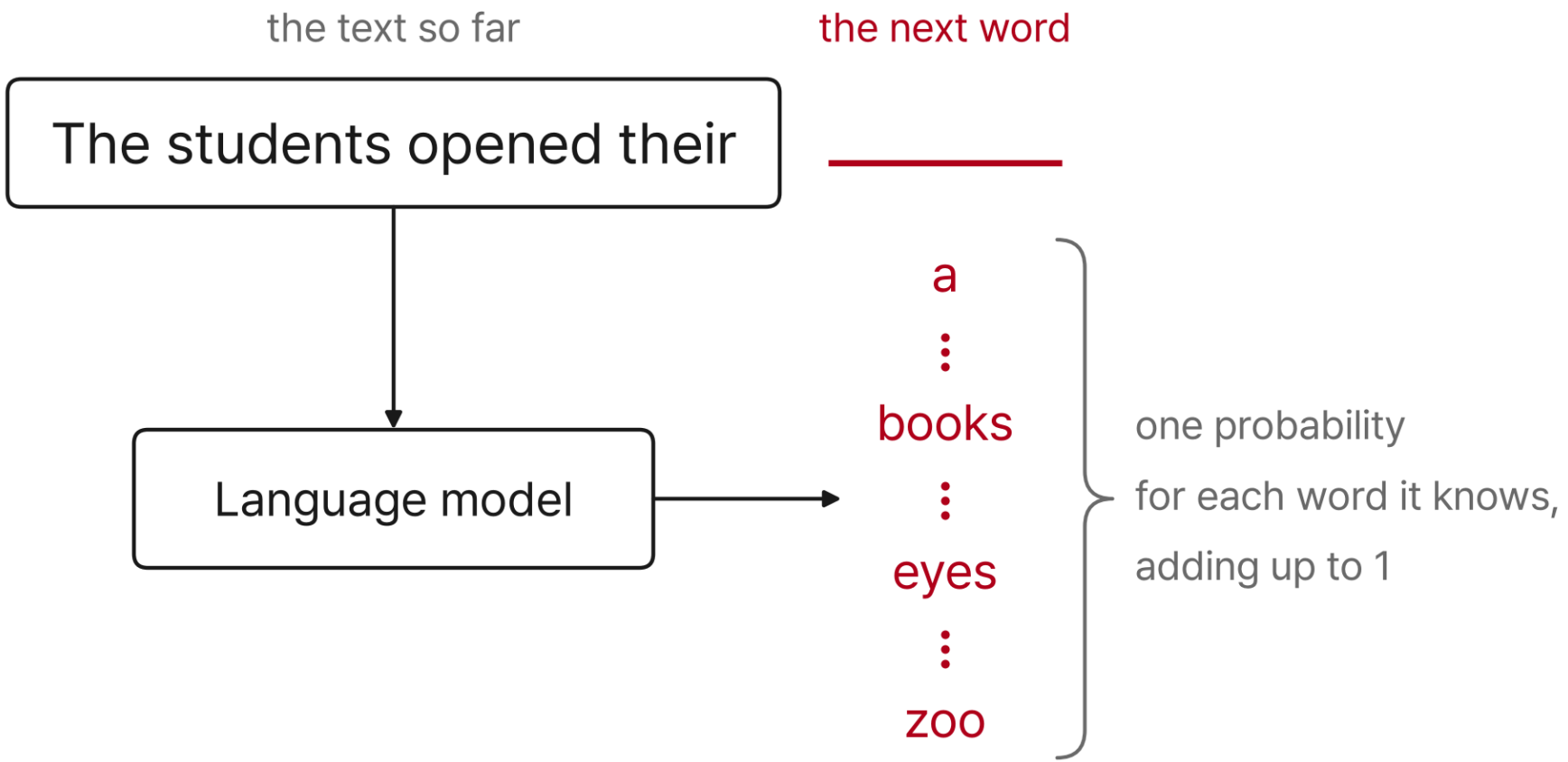
- Assignment 2: which city was this photo taken in, with a leaderboard scored every night
- Mid-point check: Friday 30 October, 22:00

Today

- Language models: predicting the next word
- A recurrent neural network that learns the next letter of a name

Language Models

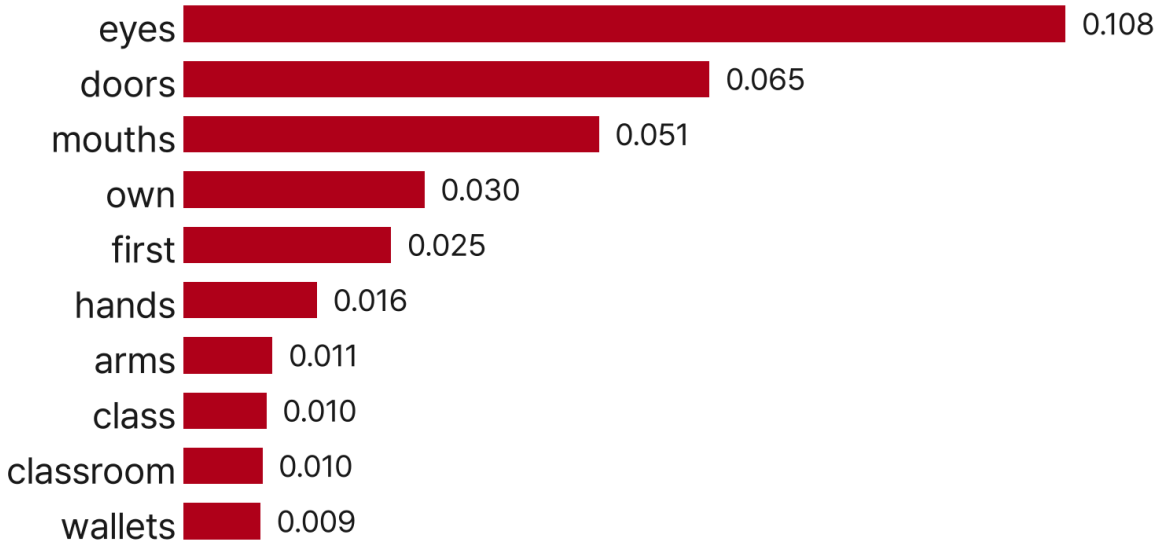
Language Model



- A model that predicts the next word from the text so far

Language Model

GPT-2 small (2019): the next word after "The students opened their"



- These 10 add up to 0.335
- The other 50,247 words share the remaining 0.665

A Classifier with One Output per Word

- Lecture 7: an image of a digit → 10 probabilities
- A language model: the text so far → one probability for each word it knows
- The same softmax and the same loss: −log of the probability given to the word that came next

Training Pairs from Text

The first sentence of "A Scandal in Bohemia", Arthur Conan Doyle (1891)

The text so far	The next word
To	Sherlock
To Sherlock	Holmes
To Sherlock Holmes	she
To Sherlock Holmes she	is
To Sherlock Holmes she is	always
To Sherlock Holmes she is always	the
To Sherlock Holmes she is always the	woman

What does a model need to know to predict the next word well?

The Next Word

Sogang University is located in _____

The Next Word

I climbed up the pear tree and picked a pear.

I climbed up the apple tree and picked _____

The Next Word

The woman walked across the street,
checking for traffic over _____

The Next Word

I went to the ocean to see the fish, turtles, seals, and _____

The Next Word

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink.

The movie was _____

The Next Word

Iroh went into the kitchen to make some tea.

Standing next to Iroh, Zuko pondered his destiny.

Zuko left the _____

The Next Word

I was thinking about the sequence that goes

1, 1, 2, 3, 5, 8, 13, 21, _____

What the Blanks Require

Blank	To fill it, you need
located in ___	facts about the world
and picked ___	the a/an rule and the word after it
traffic over ___	who "the woman" is
seals, and ___	categories and common sense
The movie was ___	the writer's opinion
Zuko left the ___	where each person is
13, 21, ___	rules and arithmetic

The Same Blanks in Three Language Models

The probability each model gives to the expected word · red: its first choice

Blank	Expected word	GPT-2 small, 2019 124M parameters, 40 GB of web text	GPT-2 XL, 2019 1.5B parameters, the same 40 GB	Qwen2.5-7B, 2024 7.6B parameters, 18 trillion tokens
located in ___	Seoul	0.011	0.064	0.118
and picked ___	an	0.025	0.752	0.817
traffic over ___	her	0.046	0.495	0.899
seals, and ___	whales, dolphins	0.121	0.209	0.254
The movie was ___	bad, boring, …	0.0057	0.021	0.091
Zuko left the ___	kitchen	0.310	0.079	0.622
13, 21, ___	34	0.018	0.199	0.785

A Detective Novel

Ilya Sutskever (co-founder of OpenAI) with Jensen Huang, NVIDIA GTC, March 2023



"The better a neural network can predict the next word in text, the more it understands it."
"... 'And that person's name is ...' Predict that word."

NVIDIA GTC 2023, Fireside Chat with Ilya Sutskever and Jensen Huang (S52092), 27:46 to 28:58

Predicting the Next Word Well

Geoffrey Hinton on 60 Minutes, CBS, 8 October 2023



"Well, it's true they're just trying to predict the next word. But if you think about it, to predict the next word you have to understand the sentences."

Learning to Predict the Next Letter

Tokens

The unit a language model reads and predicts

- Characters

  `T` `o` `␣` `S` `h` `e` `r` `l` `o` `c` `k` `␣` `H` `o` `l` `m` `e` `s` `␣` `…`

- Words, split at spaces

  `To` `Sherlock` `Holmes` `she` `is` `always` `the` `woman` `.`

- GPT-2: pieces of words, 50,257 of them (next lecture)

32,033 Names

names.txt from makemore (Andrej Karpathy), first names from US Social Security records

emma	charlotte	abigail	avery	victoria	penelope
olivia	mia	emily	sofia	madison	layla
ava	amelia	elizabeth	camila	luna	riley
isabella	harper	mila	aria	grace	zoey
sophia	evelyn	ella	scarlett	chloe	nora

- 28,829 names for training, 3,204 for validation
- 27 tokens: a to z, and "." for the start and the end of a name

Five Training Examples from "emma"

Input so far	Answer
.	e
. e	m
. e m	m
. e m m	a
. e m m a	.

- Each row: a classification into 27 tokens, with the loss −log p(answer)

More Than One Right Answer

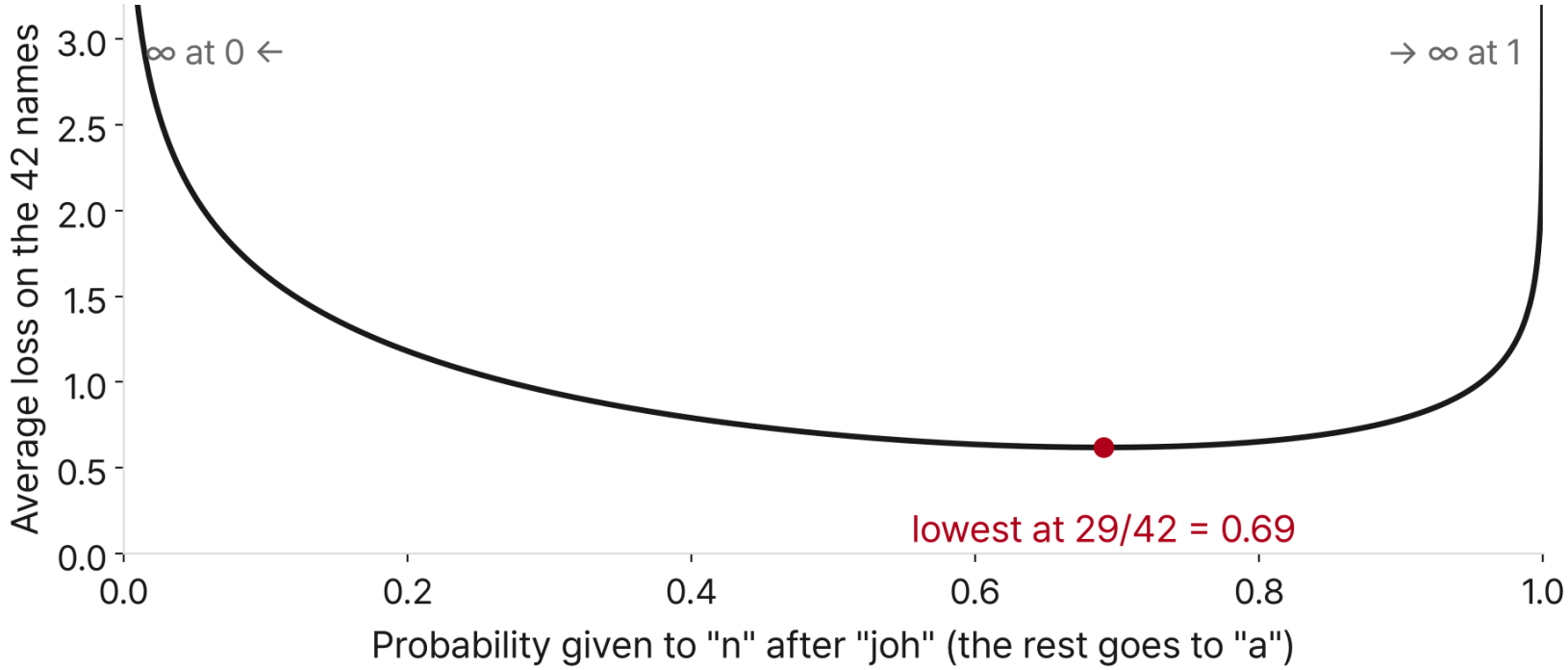
The 42 training names that start with "joh"

johaan	johannes	john	johndavid	johnmichael	johnpatrick
johan	johanny	johnae	johnell	johnni	johnpaul
johana	johany	johnanthony	johnhenry	johnnie	johnryan
johanan	johara	johnathan	johnie	johnnie	johnson
johann	johari	johnathon	johnlucas	johnniel	johnthomas
johanna	johari	johncarlos	johnluke	johnny	johnwilliam
johanne	john	johndaniel	johnmark	johnny	johny

- After "joh": "n" 29 times, "a" 13 times

The Lowest Loss

Average of −log p over the 42 names, when the model gives p to "n" and 1 − p to "a"



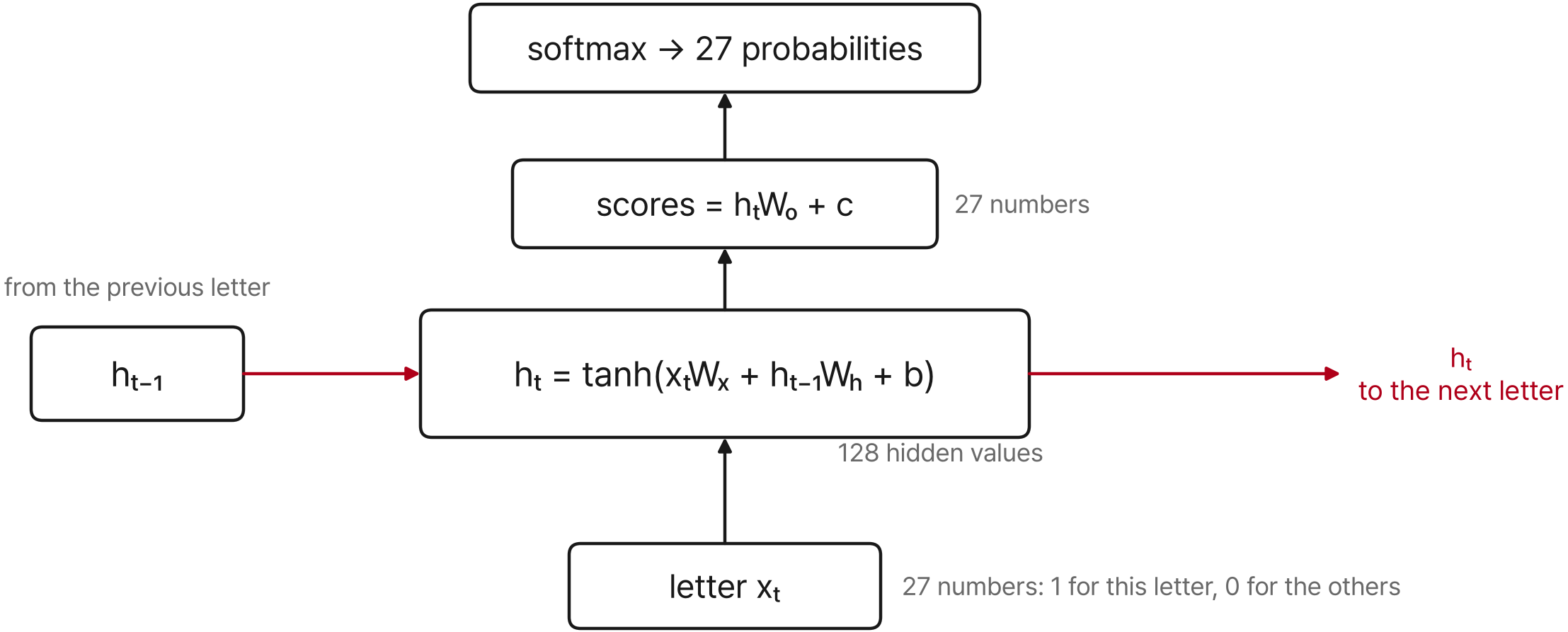
- Lowest when the probabilities equal the shares in the data: 29/42 for "n"

Names of Any Length

- An MLP takes a fixed number of inputs: 784 pixels in lecture 7
- Names have 2 to 15 letters, and the next letter can depend on all of them
- A recurrent neural network (RNN) reads one letter at a time and carries 128 hidden values forward

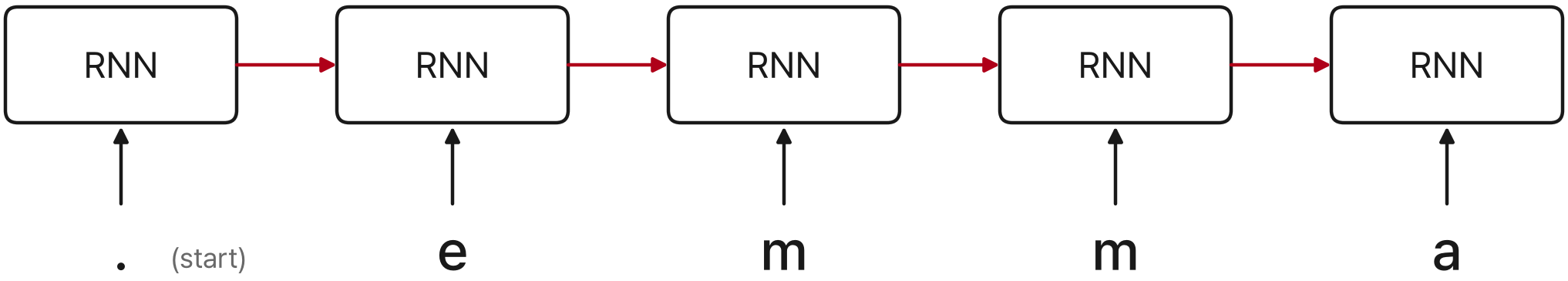
Recurrent Neural Network

Reading one letter



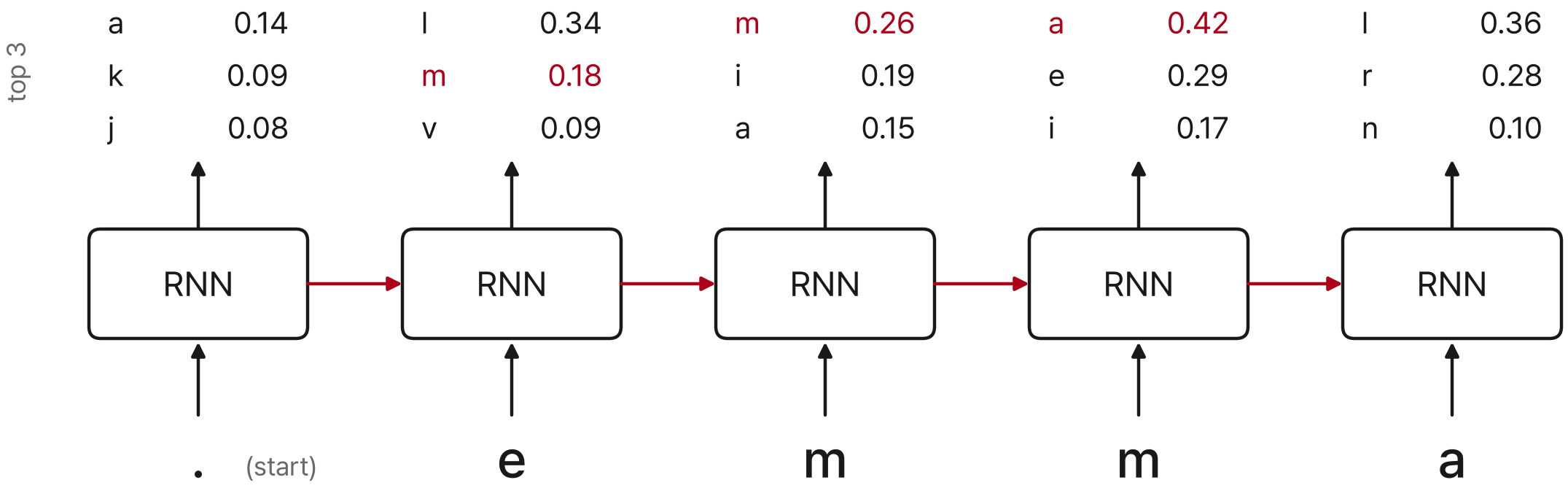
The RNN Reading "emma"

The trained model, one letter at a time



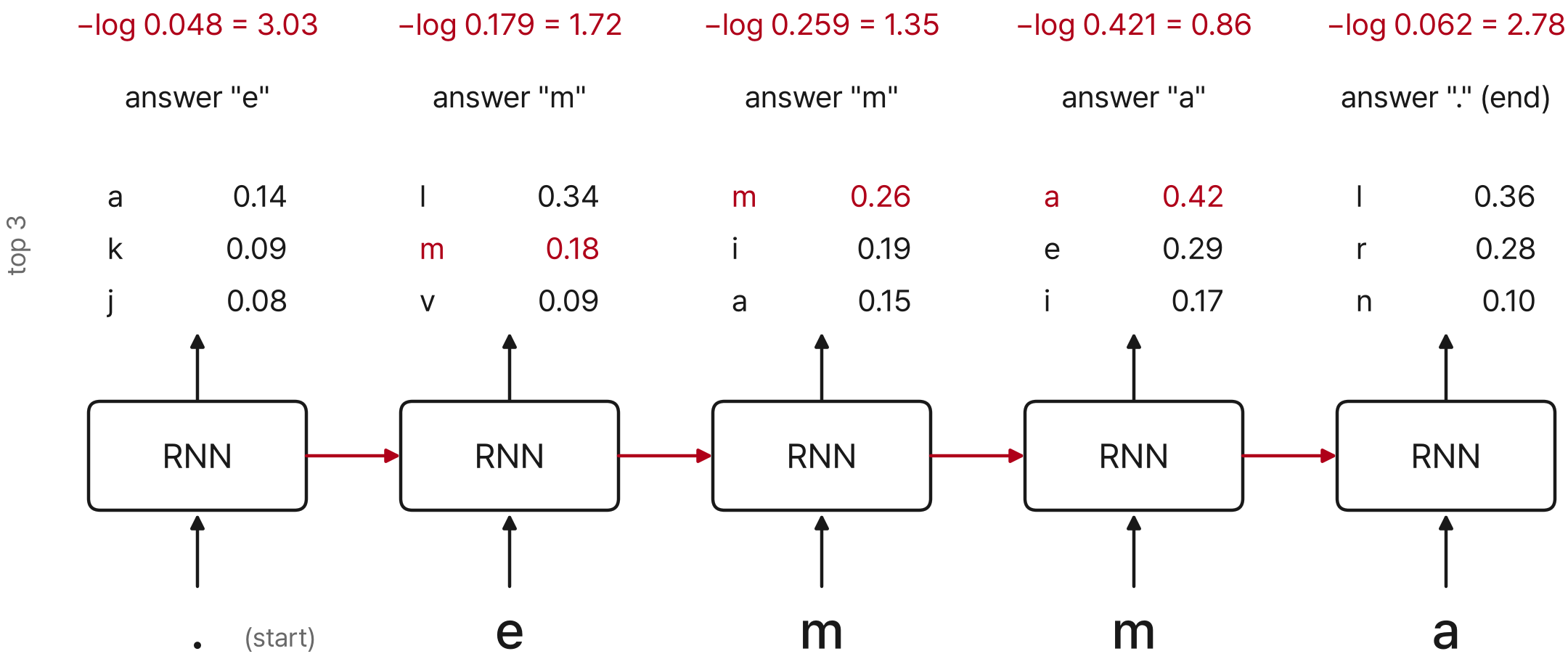
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The RNN Reading "emma"

The trained model, one letter at a time



Counting the Parameters

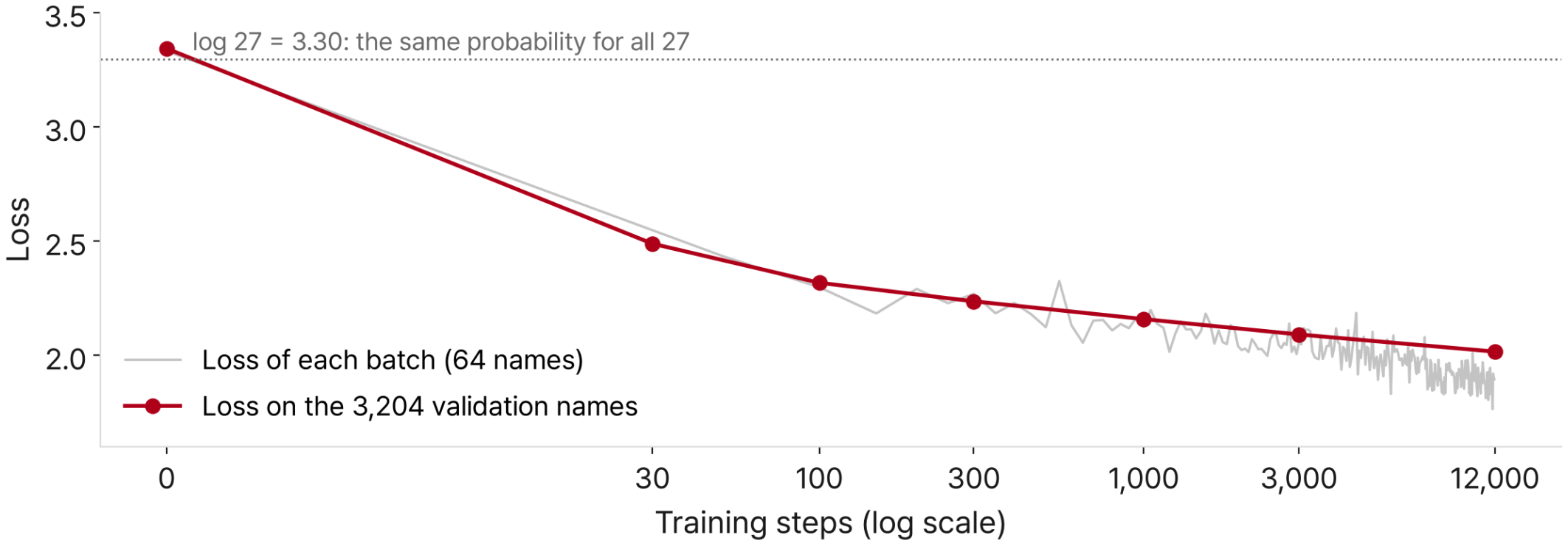
The same parameters for every letter

$$\underbrace{27 \times 128}_{W_x} + \underbrace{128 \times 128}_{W_h} + \underbrace{128}_{b} + \underbrace{128 \times 27}_{W_o} + \underbrace{27}_{c} = 23{,}451$$

- The same 23,451 numbers for a name of 3 letters or 15 letters
- Like a convolution filter in lecture 9: the same weights at every position

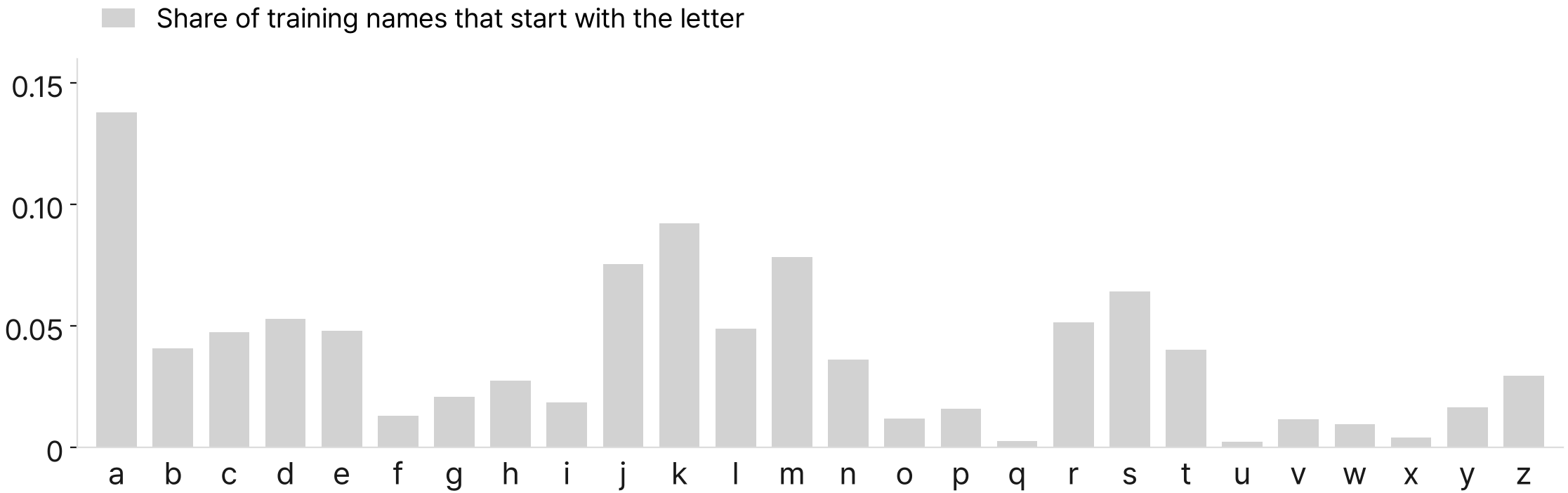
Training

12,000 steps of 64 names, about a minute and a half on a CPU



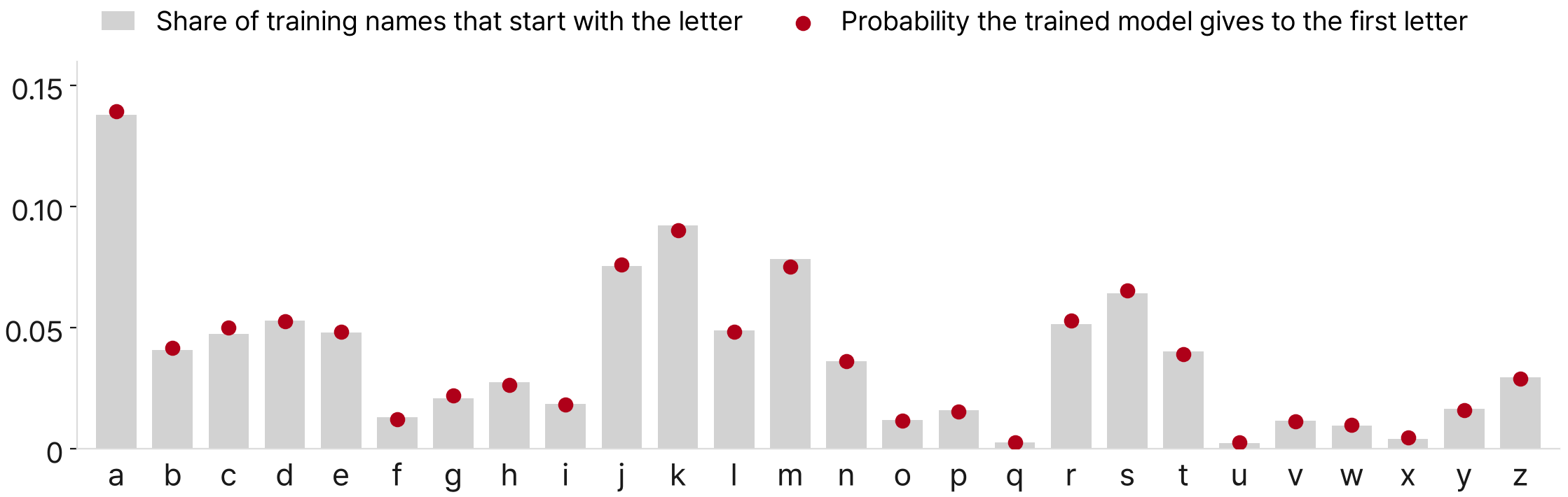
The First Letter

The first letter of the training names, and the trained RNN after "."



The First Letter

The first letter of the training names, and the trained RNN after "."



- After "joh", backed by only 42 names, the trained model gives "n" 0.38, not 0.69

Generating a Name

The trained RNN after ".": "a" is picked with probability 0.139

- Compute the probabilities of the next letter from the letters so far
- Pick one at random with those probabilities, and add it to the name
- Feed it back as the next input, until "." is picked

Names During Training

Names 1 to 6 of seed 1 in the demo: the same random numbers for every step

Steps	Validation loss	Sampled names
0	3.343	zeykfnzatjpa, lwryogqkenxrprgszb, dkrsofeva, zablz, nz, pdy
100	2.318	zayiaht, jrkyman, alirh, zadiy, kzasan, lay
1,000	2.159	zayeidu, josye, amirh, zadey, kyarah, lay
12,000	2.016	zaylahn, josya, alivia, zadiy, kyan, layanna

Names RNN

jdasam.github.io/mas1004/demos/lecture13-names-rnn



- Generate names; click a bar to choose the next letter · Score your own name

Scoring a Name

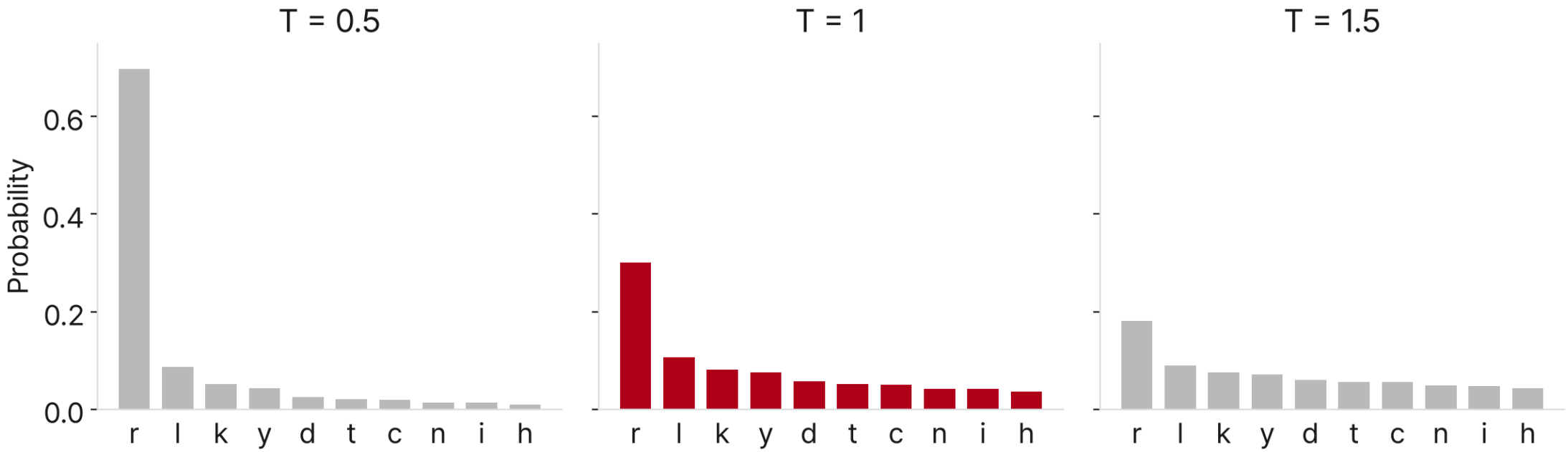
The model after 12,000 steps

Position	The model has seen	Next character	RNN probability	−log p	
1	.	d	0.053	2.94	
2	. d	a	0.417	0.87	
3	. d a	s	0.027	3.61	
4	. d a s	a	0.048	3.03	
5	. d a s a	e	0.019	3.98	
6	. d a s a e	m	0.0094	4.67	least likely
7	. d a s a e m	.	0.903	0.10	

- Average −log p: 2.74 for dasaem, against 2.016 over the validation names

Sampling and Temperature

Next letter after "ma": softmax(scores / T)




```
T = 0.5: zaylah, hensley, alexandra, zahar, juan, kayanna, renna, kailynn (5 of 20 are new)
T = 1.5: zayleny, kwkyn, blisob, zachy, kyaq, may, thiocl, madeyvia (16 of 20 are new)
```

New Names

1,000 names sampled from the trained model at T = 1 (the demo, seed 1)

- 218 are training names, 15 are validation names it never saw
- 764 are not in the dataset at all

```
zaylahn, josya, zadiy, shephia, doniel, niemi, shanvie, lanets, valla, eddiz, oley, kymbert
```

Coming Up

- Friday 30 October, 22:00: Assignment 2 mid-point check
- Tuesday 3 November: the mid-point leaderboard, and a few of you share one change that helped
- Next lecture: large language models